

Quadratic functions - Set 3 - Solutions

Specification MM1.3, M4.5, M4.11-M4.14.

Answer key

Question	Answer	Topic
1	D	Quadratic functions
2	H	Quadratic functions

Worked solutions

Question 1 QUA-22 · Quadratic functions · spec MM1.3

The equation $x^2 + ax + b = 0$ has roots 3 and -5 . The equation $x^2 + bx + a = 0$ has roots p and q . Find $p^2 + q^2$.

Answer: D

Sum of roots $3 + (-5) = -a$, so $a = 2$; product $-15 = b$. Then $x^2 - 15x + 2 = 0$ has $p + q = 15$ and $pq = 2$, so $p^2 + q^2 = 15^2 - 2(2) = 221$.

Question 2 X2-01 · Quadratic functions · spec MM1.3

p and q are real numbers. Which of the following statements is/are true?

- I If $p^2 < 4q$ then $x^2 + px + q > 0$ for all real x .
- II If $x^2 + px + q > 0$ for all real x then $q > 0$.
- III If $q < 0$ then $x^2 + px + q = 0$ has two real roots of opposite sign.

Answer: H

I: negative discriminant and positive leading coefficient. II: put $x = 0$. III: the product of the roots is $q < 0$ (and the discriminant $p^2 - 4q > 0$), so the roots are real with opposite signs.